



2025-4 - klausur

Machine learning (Technische Universität München)



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E0301

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Maschinelles Lernen

Exam: IN2064 / Retake

Date: Monday 14th April, 2025

Examiner: Prof. Dr. Stefan Leutenegger
Dr. Leo Schwinn

Time: 13:30 – 15:30

P 1

P 2

P 3

P 4

P 5

P 6

P 7

P 8

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Working instructions

- This exam consists of **20 pages** with a total of **8 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 47 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

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Problem 1 Decision Trees (5 credits)

An online vendor is currently selling two kinds of item, A and B , and soon plans to offer a new item C . They asked a set of eight customers if they would buy item C and if they bought items A and/or B in the past:

Customer No.	x_A ("Bought A ")	x_B ("Bought B ")	y ("Wants C ")
1	False	False	False
2	False	False	False
3	False	True	True
4	False	True	False
5	True	False	True
6	True	False	True
7	True	True	False
8	True	True	True

With this data, you are tasked to build a decision tree predicting whether a customer might want item C , based on their past preferences for items A and B .

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a) Calculate the Gini index $i_G(y)$ of the class labels y . **Justify** your answers.

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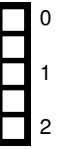
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b) Calculate which feature is best for the first split, using the Gini index i_G as an impurity measure. **Justify** your answers.



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c) Assume now that, instead of just a single feature, you can split based on the result of an arbitrary Boolean function of both features, $f: \{\text{True}, \text{False}\}^2 \rightarrow \{\text{True}, \text{False}\}$, $(x_A, x_B) \mapsto f(x_A, x_B)$. Write down a particular criterion f under which the purity after the first split becomes even better compared to your split chosen in (b). Compute the new resulting purity gain. **Justify** your answers.





Problem 2 Linear Regression (7 credits)

Consider the standard linear regression model from the lecture (here we **ignore the bias term**), where the output is modeled as

$$f(x) = \mathbf{w}^\top \phi(x) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (2.1)$$

with $\phi(x) \in \mathbb{R}^D$ being a feature map and $\mathbf{w} \in \mathbb{R}^D$ the weight vector. The training data is represented as $\mathcal{D} := \{x_i, y_i\}_{i=0}^N$ where the targets $y_i \in \mathbb{R}$.

We now propose the following two **variants** of this model:

Variant 1: Noisy Features

Instead of a deterministic feature map $\phi(x)$, we add independent Gaussian noise to the features. That is, each input is mapped using $\tilde{\phi}$ as:

$$\tilde{\phi}(x) := \phi(x) + \mu_i, \quad \mu_i \sim \mathcal{N}(0, \Sigma) \quad (2.2)$$

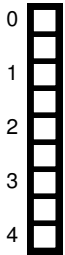
Finally, minimize the **expected mean-squared loss** over this noise, i.e., we minimize:

$$\mathcal{L} = \mathbb{E}_{\mu_1, \dots, \mu_N} \sum_{i=1}^N (\mathbf{w}^\top (\phi(x_i) + \mu_i) - y_i)^2 \quad (2.3)$$

Variant 2: Data-Dependent Noise

Here we assume that the noise variance depends on the input. The model becomes:

$$f(x) = \mathbf{w}^\top \phi(x) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma(x_i)^2) \quad (2.4)$$



a) **Prove** that the **Variant - 1** is equivalent to using $\mathbf{w}^\top \Sigma \mathbf{w}$ as the regularization term. **Derive** the optimal weight assuming $\Sigma = \frac{\sigma_1^2}{N} \mathbb{I}$.

Hint 1: Use linearity of $\mathbb{E}$ and $\mathbb{E}[\mu_i] = \mathbf{0}$; $\mathbb{E}[\mu_i \mu_i^\top] = \Sigma$.

Hint 2: The solution to Ridge regression is $\mathbf{w}^* = (\Phi^\top \Phi + \lambda \mathbb{I})^{-1} (\Phi^\top \mathbf{Y})$, where $\Phi \in \mathbb{R}^{N \times D}$ is the design matrix.





b) Suppose that the matrix $\Phi^T \Phi$ is not invertible (i.e., the design matrix is rank-deficient). Which of the two variants is more likely to help address this issue? **Justify** your answer. You can either use insights from the solution to previous subproblem (a) or answer **independently**.

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c) You observe that the standard regression model performs worse on inputs from a specific subset of data points, indicating a potential bias in data acquisition. Which of the two variants is more likely to help address this issue? **Justify** your answer. You can either use insights from the solution to previous subproblem (a) or answer **independently**.

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Problem 3 Linear Classification (4 credits)

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a) Describe a Naive Bayes model for the following dataset and fit it to the data in the table. Dashes (–) denote missing data.

Speed x_1	Color x_2	Class y
40	red	1
–	green	0
80	–	1
30	blue	1
60	blue	0
50	green	1

Specify all required distributions and compute their parameters. If your solution uses Normal distributions, assume fixed variances of $\sigma^2 = 10$.

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b) Which class does your model assign to the new data point (–, –). The dash (–) denotes missing data. **Justify** your answer.





Problem 4 Optimization (6 credits)

A common mathematical operation is the log-sum-exp operation $f : \mathbb{R} \rightarrow \mathbb{R}$ of a linear projection defined as

$$f(x) = \log \sum_{i=1}^n \exp(a_i x + b_i). \quad (4.1)$$

Proof that f is convex with respect to x .

Hint 1: You might **not** want to use convexity-preserving operations.

Hint 2: The variance of a categorical distribution $p(c) = \pi_c$ for classes $c \in \{1, \dots, C\}$ is positive, i.e.,

$$\text{Var}_{\pi}(c) = \sum_{c=1}^C \pi_c c^2 - \left(\sum_{c=1}^C \pi_c c \right)^2 \geq 0. \quad (4.2)$$





Problem 5 Deep Learning (8 credits)

Consider a fully connected neural network which maps two inputs $x_1, x_2 \in \mathbb{R}$ to a single output $y \in \mathbb{R}$ as follows:

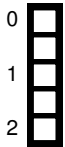
$$h_1 = \max(0, w_1 x_1 + w_2 x_2 + b_1) \quad (5.1)$$

$$h_2 = \max(0, w_3 x_1 + w_4 x_2 + b_2) \quad (5.2)$$

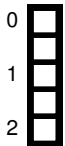
$$y = w_5 h_1 + w_6 h_2 \quad (5.3)$$

and is trained by optimizing the Mean Squared Error (MSE), where $\hat{y}$ is the target value:

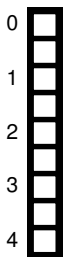
$$L = \frac{1}{2}(y - \hat{y})^2 \quad (5.4)$$



a) **Draw** a computation graph for the neural network and **derive** an expression for the gradient of the loss w.r.t. weight w_1 .



b) **Derive** condition on w_1, w_2, b_1 that guarantees $h_1 = 0$ for all $(x_1, x_2) \in \mathbb{R}^2$. Explain why this might be problematic for training. **Justify** your answer.



c) Now assume $w_5, w_6 = 1$, $b_1, b_2 = 0$ and $w_1, w_2, w_3, w_4 \in \{-1, 1\}$, show that this neural network can model the XOR function:

$$\text{XOR}(x_1, x_2) = \begin{cases} 0, & \text{if } (x_1, x_2) \in \{(0, 0), (1, 1)\} \\ 1, & \text{if } (x_1, x_2) \in \{(0, 1), (1, 0)\}. \end{cases}$$

Find the right configuration for w_1, w_2, w_3, w_4 to solve the XOR function and show that it does. **Justify** your answer

Hint: You can start by interpreting the meaning of h_1, h_2 .





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Problem 6 Kernels (5 credits)

A vector of m nonnegative integer entries $\mathbf{x} = (x_1, \dots, x_m)^T \in \mathbb{N}_0^m$ is said to represent a *histogram of N samples over m bins* (denoted by $\mathbf{x} \in \text{Hist}(N, m) \subset \mathbb{N}_0^m$) if its entries also fulfill $\sum_{i=1}^m x_i = N$. Consider the function

$$k_{N,m} : \text{Hist}(N, m) \times \text{Hist}(N, m) \rightarrow \mathbb{R}, \quad k_{N,m}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^m \min(x_i, y_i).$$

- 0 ☐ a) For a fixed N , for which cases of input pairs does $k_{N,m}$ take on its minimum and maximum output values? More generally, given any two histograms $\mathbf{x}, \mathbf{y}$, how would you interpret the result $k_{N,m}(\mathbf{x}, \mathbf{y})$? **Justify** your answers.

- 0 ☐ b) Prove that $k_{N,m}$ is a kernel for all integers $N > 0$ and $m > 0$. Write down a suitable feature map ϕ first, making sure to argue why your choice of ϕ is well-defined.
- 1 ☐ **Hint 1:** Consider a ϕ which maps to a vector space of matrices with binary entries.
- 2 ☐ **Hint 2:** You may assume that, for all $K, L \in \mathbb{N}$ and $A, B \in \mathbb{R}^{K \times L}$, the Frobenius inner product $\langle A, B \rangle_F = \sum_{k=1}^K \sum_{l=1}^L (A)_{kl}(B)_{kl}$ is indeed an inner product.
- 3 ☐





c) **Justify** why $k_{N,m}$ (as defined above) is not a kernel in the following case:

- We keep requiring $\sum_{i=0}^m x_i = N$, but allow entries x_i to be negative.

Hint: Consider Mercer's condition.





Problem 7 Principal Component Analysis (6 credits)

Consider a dataset as in 7.1.

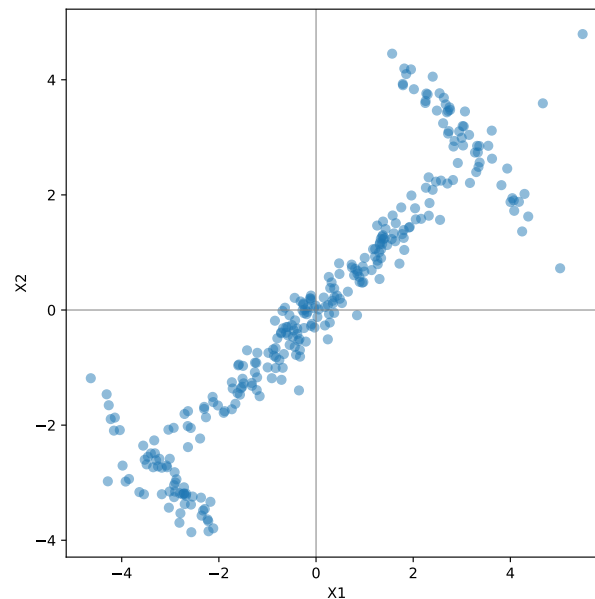


Figure 7.1: Toy dataset

- 0 ☐ a) Draw (approximately) the first two principal components of the dataset in 7.1 and explain how you obtained them. **Justify** your answer.

1 ☐

- 0 ☐ b) Can you find another principal component? If yes, draw it. If no, explain why not. **Justify** your answer.

1 ☐

- 0 ☐ c) Now, consider the following linear autoencoder with a single hidden layer from 7.2. We have no biases and the activation function is the identity function.

1 ☐

- 2 ☐ If you use maximum likelihood optimization for training, can you obtain a set of weights that perfectly reconstruct the input? If yes, specify the weights. If no, explain why not. **Justify** your answer.



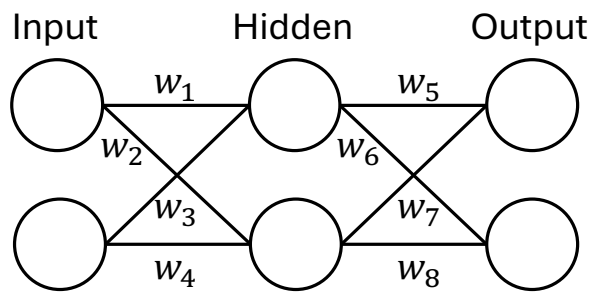
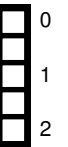


Figure 7.2: Autoencoder



d) Given the normalized (i.e. 2-norm is 1) first principal component $\vec{v}^{(1)} = [v_1, v_2]$ specify the weights w_1, w_2, w_3, w_4 of the autoencoder to simulate the projection of the data onto the first principal component? **Justify** your answer.





Problem 8 Clustering (6 credits)

Gaussian Mixture Models (GMMs) model the distribution over data as:

$$P(x) = \sum_{k=1}^M \pi_k \mathcal{N}(x; \mu_k, \Sigma_k) \quad (8.1)$$

Recall that each iteration of the EM algorithm for GMMs consists of two steps:

1. The **E-Step** computes the responsibilities: $P(z_{ik} = 1 | x_i, \pi, \mu, \Sigma)$.
2. The **M-Step** updates the GMM parameters: π, μ, Σ .

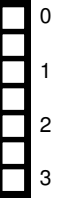
Assume that the covariance matrices are fixed during training and set to eq. (8.2) for every Gaussian in the mixture model. Recall that as the hyperparameter $\sigma \rightarrow 0$, the EM algorithm reduces to Lloyd's algorithm with $L2$ distance. Now suppose we wish to run Lloyd's algorithm using the Mahalanobis distance defined as $d_{\Sigma}(x_i, x_j) := \sqrt{(x_i - x_j)^{\top} \Sigma^{-1} (x_i - x_j)}$, for $\Sigma \succ 0$ (symmetric positive definite), as a **special** case of EM algorithm.

$$\Sigma_k = \sigma^2 \mathbb{I} \quad (8.2)$$



a) Write down the **E-Step** and **only** the update of μ in the **M-Step** of the EM algorithm.





b) How should we change the parameterization of the covariance matrices Σ_k from eq. (8.2) such that as the hyperparameter $\sigma \rightarrow 0$ the **E-Step** of EM algorithm converges to the cluster assignment of Lloyd's algorithm with $d_{\Sigma}(\cdot, \cdot)$ as the distance metric? **Justify** your answer.





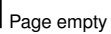
- 0 ☐
- 1 ☐
- 2 ☐

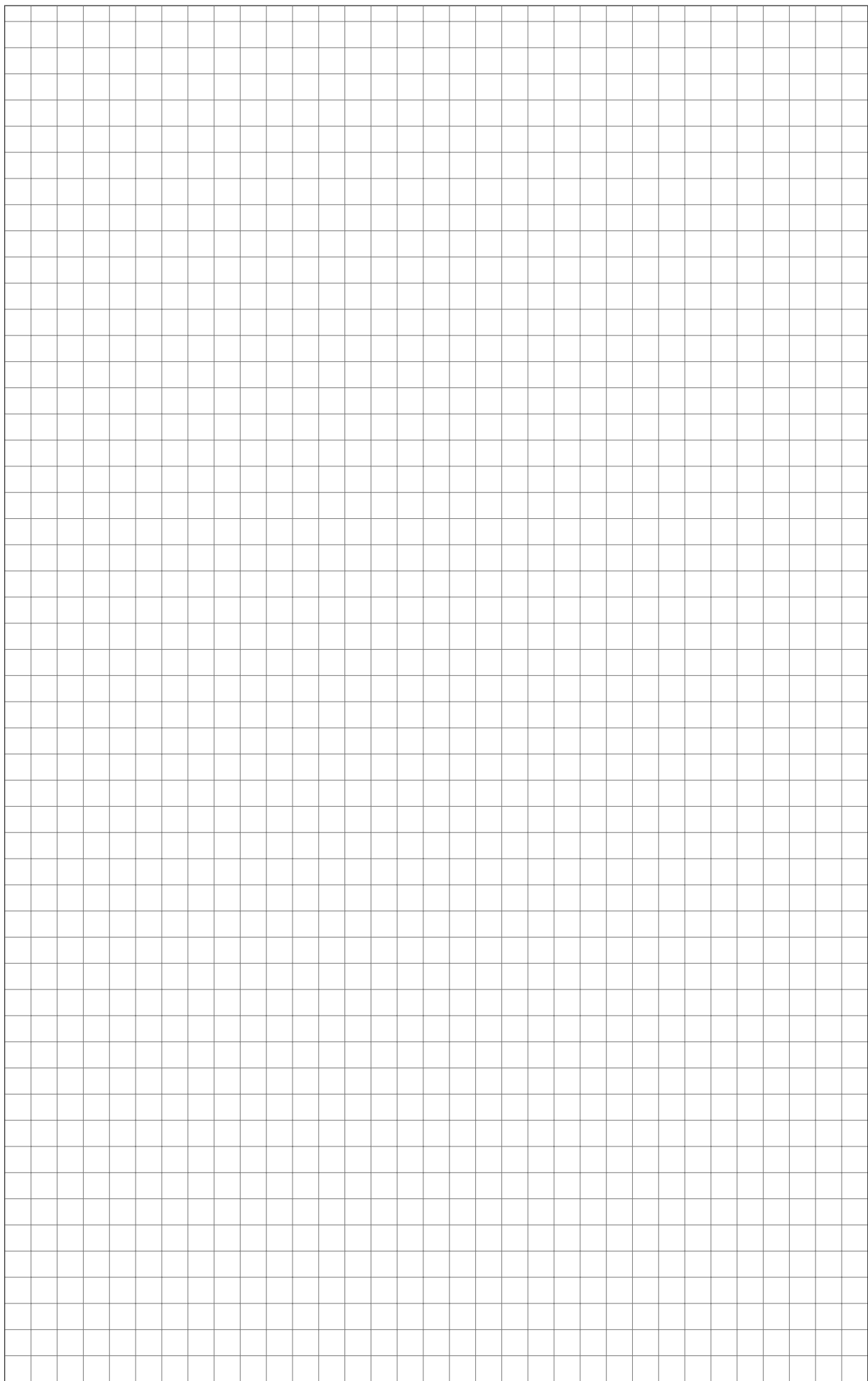
c) After performing the **E-Step** as per subproblem (b), write the **M-Step** only updating the parameter μ . Does it corresponds to the centroid update step of the Lloyd's algorithm with $d_{\Sigma}(\cdot, \cdot)$ as the distance metric ? **Justify** your answer by arguing about the centroid update for Mahalanobis distance in the Lloyd's algorithm.

Hint 1: You can solve this **independent** of (b) by assuming that **E-Step** in (b) leads to hard cluster assignment as per the Lloyd's algorithm.

Hint 2: The centroid of the points with respect to the distance $d_{\Sigma}(\cdot, \cdot)$ is independent of Σ .







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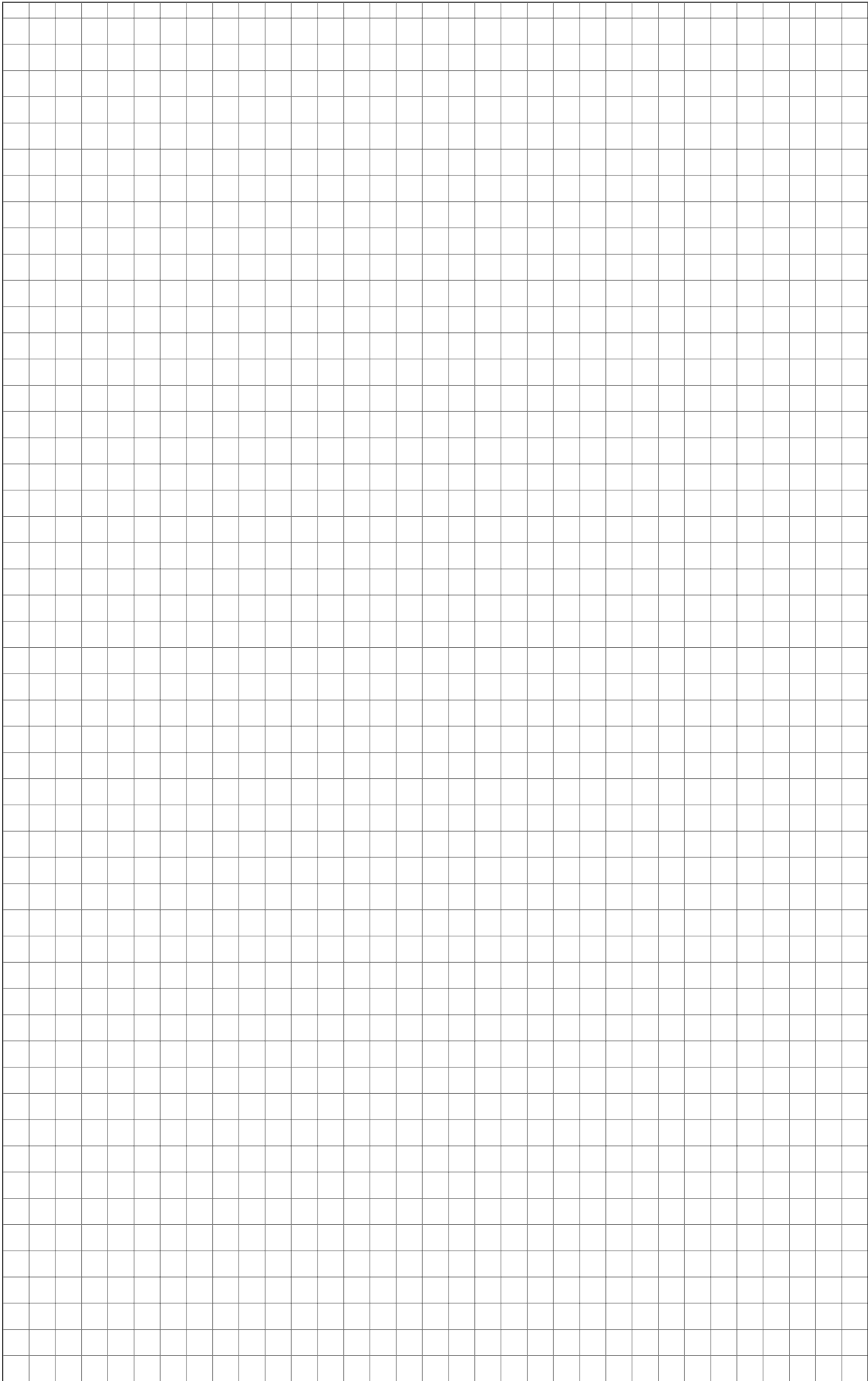
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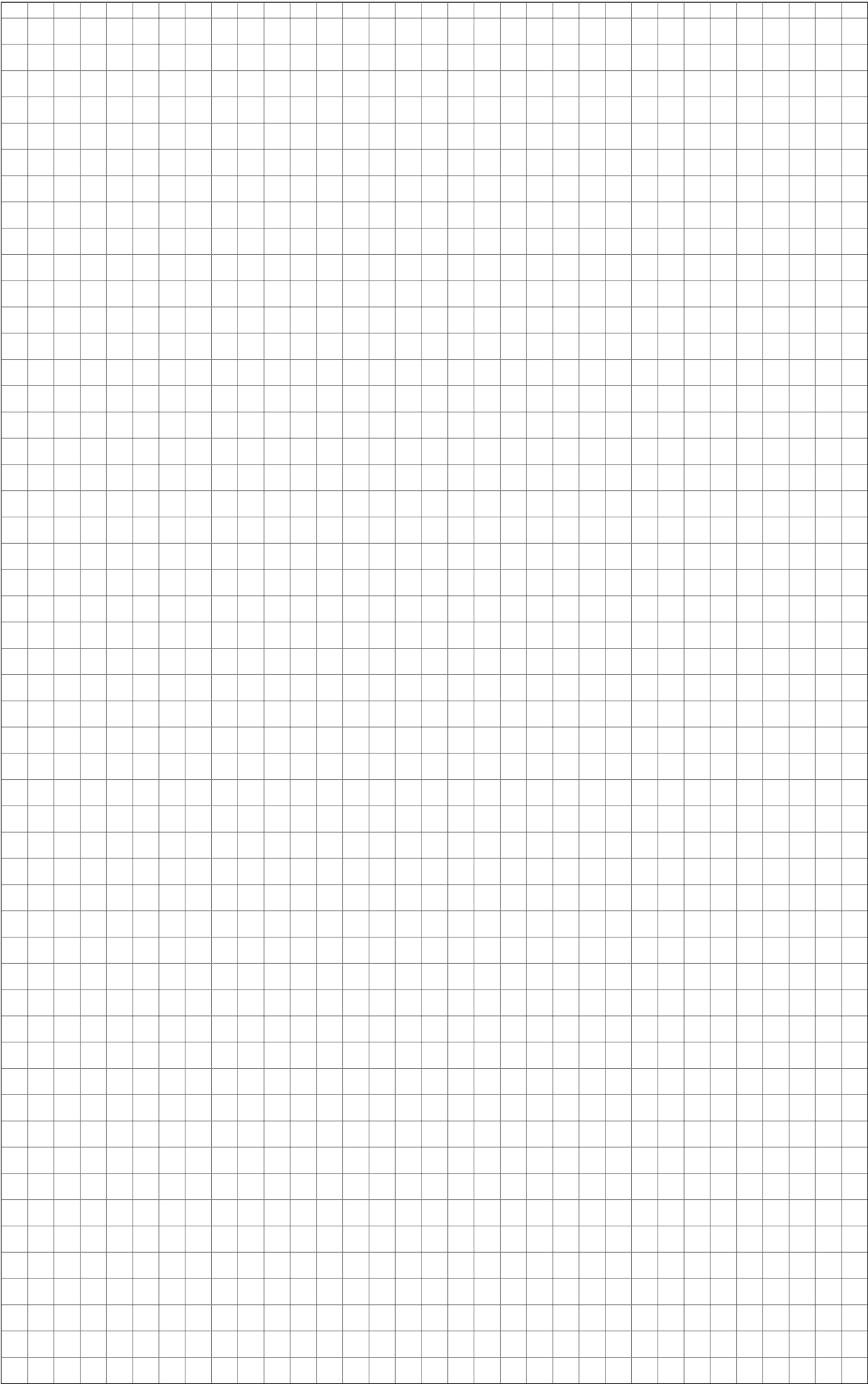




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